DAY 1 · MODULE 1 · 25 MIN

Azure AI Landscape

Where MAF, Foundry, and Copilot Studio fit

Why this module

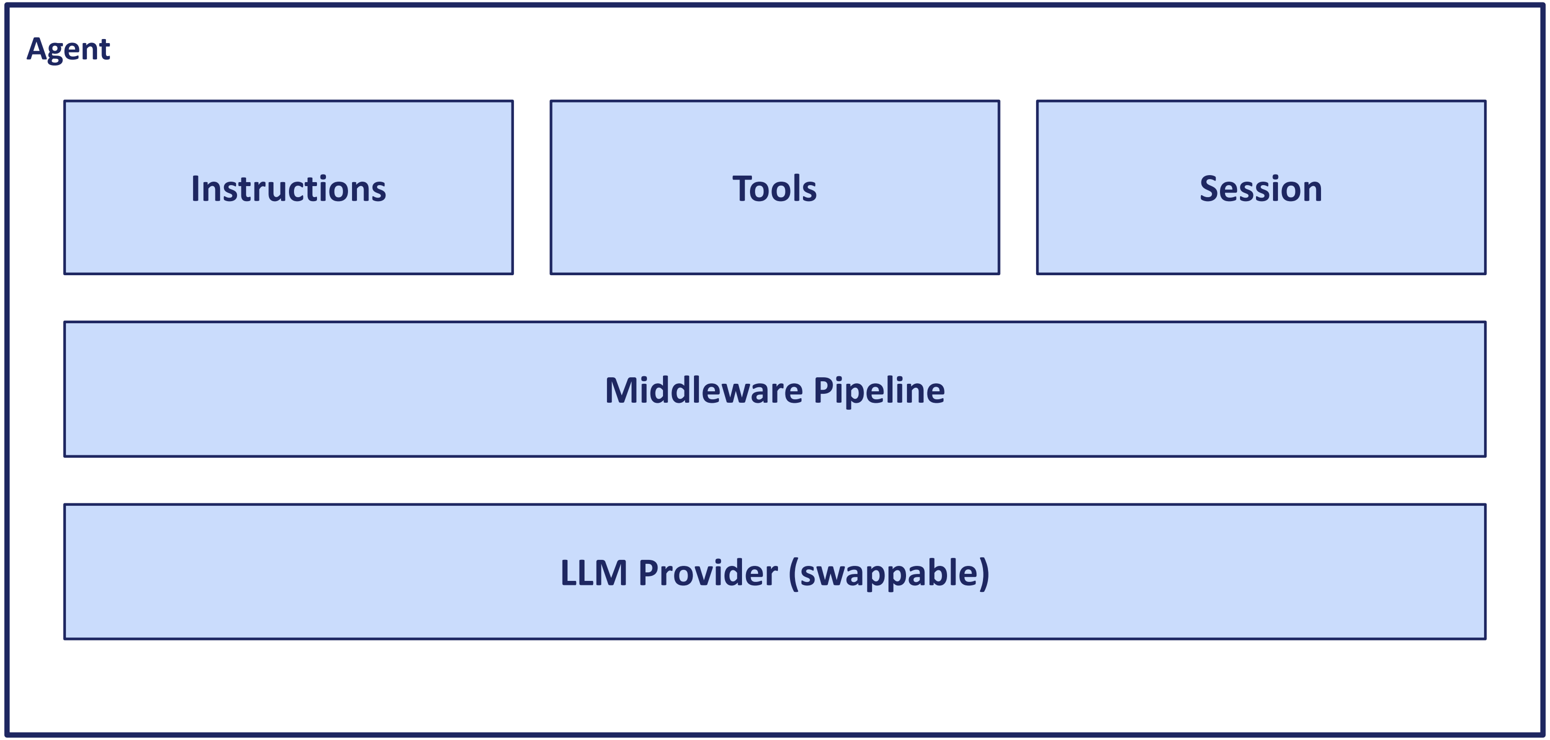
By the end of this workshop you'll be building agents. Before writing code, you need to know which surface to build on — and when. Today's decisions have outsized impact on cost, control, and portability.

What is an agent?

Raw LLM call	Agent
Full control over every API parameter	Opinionated abstractions that handle common patterns
No memory, no tools, no identity	Persistent identity + tools + session + middleware
You wire up state, tool dispatch, retry logic	Framework handles the loop
Tightly coupled to one provider	Swap providers without changing app code

An agent wraps an LLM with the structure needed to build real applications — persistent identity, instructions, tools, memory, and a runtime loop that orchestrates it all.

What an agent adds



The rest of the workshop works with one or more of these layers directly.

Diagram: Microsoft Learn · aka.ms/agent-framework/journey/from-llms-to-agents

D E M O

~4 min

What an agent adds — live diff

Two terminal panes, same question. Left: raw LLM call — no tool, no session, no memory. Right: MAF Agent with get_current_time tool + explicit session. Left says it can't get the time and forgets Turn 1; right calls the tool, gets the real time, and recalls the earlier question. Makes the 'agent wraps LLM' diagram tangible.

Runbook: demos/day1/module-1-demo-2-agent-diff.md

Three surfaces you can build on

Surface	Audience	Sweet spot
Copilot Studio	Makers, business analysts	Low-code agents for M365, Teams, Power Platform
Microsoft Foundry	Developers	Full model + agent platform on Azure
Microsoft Agent Framework (MAF)	Developers	The SDK your app code uses to build agents on top of Foundry

Same three ingredients across every surface

Copilot Studio	Foundry Agent Service	MAF + containers
SaaS · low-code	PaaS · managed runtime	IaaS · you own everything

All three need:

Model | **Instructions** | **Tools**

Same ingredients everywhere. What changes is how much code you write and who owns the runtime.

A decision framework, visualized

More control, more code — **Faster to build, less customization**

IaaS	PaaS	SaaS
Containers + OSS frameworks	**Foundry Agent Service + MAF**	**Copilot Studio**
You own everything. Not covered.	Managed runtime; you own the agent code. This workshop.	Portal-first, low-code. Not covered.

This workshop lives in PaaS territory. You write real code; you don't run the infrastructure.

When is an agent the right answer?

Not every LLM problem needs an agent. Match complexity to need.

Direct LLM call

"Summarize this document."

No tools · no iteration · cheap · easy to eval

Single agent with tools

"Answer with sources; open a ticket if needed."

Iteration · tool calls · harder to eval

Multi-agent workflow

"Plan → retrieve → verify → act."

Multiple roles · costly · new failure modes

Reach for the leftmost pattern that actually solves your problem

What Foundry gives you

- Model deployments (frontier and open models)
- Playgrounds — chat, agent, evaluation
- Foundry Agent Service — runtime for Prompt agents (portal-authored, no code) and Hosted agents (your code, containerized, Foundry-run)
- Responses API — the single model + tools entry point; call it from your own code running anywhere
- Foundry Toolbox — curated tool/connector catalog, exposed via MCP
- Foundry IQ — enterprise knowledge and grounding layer
- Evaluators, tracing, and safety

What MAF gives you

- One SDK for authoring agents — Python and C#
- One vocabulary: agents, sessions, tools, runs
- First-class streaming, memory, structured outputs
- Multi-agent orchestration primitives
- Consumes Foundry Toolbox and MCP servers
- The successor to Semantic Kernel and AutoGen patterns (both out of scope)

What we will and won't cover

Covered

- Foundry (portal, deployments, connections)
- MAF — two hosting families (Foundry-hosted and self-hosted), covered in Module 6
- Toolbox and Foundry IQ
- MCP consumption and (stretch) authoring
- Evaluation at every layer
- Production concerns: observability, identity, cost

Out of scope

- Copilot Studio (different audience)
- Semantic Kernel (MAF is the forward direction)
- AutoGen (research-lineage predecessor)
- Third-party frameworks (LangChain, CrewAI, etc.)
- Model fine-tuning, distillation, training
- On-prem / air-gapped deployments

Roadmap — how the week hangs together

Day	Theme
1	Foundations: the stack; three hosting options; first working agents
2	Grounding (Foundry IQ + custom RAG) and tools in depth
3	Single-agent depth: memory, streaming, structured outputs, MCP
4	Multi-agent patterns + evaluation as a first-class activity
5	Production concerns; capstone project kickoff

Takeaways

- Three surfaces target different audiences: Copilot Studio, Foundry, MAF.
- This workshop lives at Foundry + MAF.
- MAF is Microsoft's forward direction for agent SDKs; SK and AutoGen are not.

Next: A working tour of the Foundry portal — projects, models, deployments, Toolbox, Foundry IQ.